



Generative adversarial network and texture features applied to automatic glaucoma detection

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ABSTRACT

Glaucoma is a neurodegenerative disease that has a multifactorial etiology. The main characteristic of this illness is the progressive lesion of the optic nerve. This disease is chronic and causes permanent blindness at an advanced stage. Early diagnosis is essential to ensure a favorable prognosis and improve the patient's quality of life. Digital Image Processing together with computational techniques of machine learning allow the creation of methods for automatic detection of glaucoma. In this context, this work aims at the early diagnosis of glaucoma through a Generative Adversarial Network allied to texture attributes defined from indexes of taxonomic diversity. The method we propose can be divided into: (i) image acquisition through the RIM-ONE and Drishti-GS public databases; (ii) training of a conditional Generative Adversarial Network for segmentation of the optical discs into retinal images; (iii) pre-processing through enhancement and hole fill-in techniques; (iv) extraction of texture attributes using the index of taxonomic diversity; and (v) validation of the proposal through three classifiers evaluated according to four performance metrics. The results are promising and indicate that the method is robust, initially reaching 77.9% accuracy. However, as we apply improvements and adjustments in the method employed, we reach 100% accuracy and a ROC curve of 1. Therefore, we propose a second opinion on the diagnosis of glaucoma, assisting the specialist precisely.

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1. Introduction

Glaucoma is a neurodegenerative disease that promotes progressive damage to the optic nerve, resulting in visual field loss [1]. As it is a chronic and asymptomatic disease, early diagnosis is essential to avoid partial or total loss of vision.

Regardless of population and social status, glaucoma is the second largest cause of blindness in the world [2,3], preceded only by cataracts. The loss of vision caused by cataracts can be reversible through surgery, whereas glaucoma causes permanent blindness. Among the factors for developing glaucoma, elevation of intraocular pressure (IOP) is the most common and primary cause of this disease. This elevation occurs due to obstruction of the drainage channel of the aqueous humor liquid, resulting in numerous ocular diseases in cells of the retina and the optic nerve.

The death of optic nerve cells (the axons) affects the increase of nerve excavation, which is the emergence of voids in the optic nerve, and, consequently, the impairment of the visual field. In

Figs. 1 and 2, we present examples of excavation (black markings) of the optic disc (OD) region (green markings) for healthy and glaucomatous images.

Based on Figs. 1 and 2, we can see that images displaying glaucoma (Figs. 1(a) and 2(a)) present more heterogeneous regions, that is, a greater variation of intensity. This same property is not so evident when we analyze the histograms of healthy images (Figs. 1(b) and 2(b)). From this, we look at the diversity of information provided by texture analysis. Histograms were calculated from the OD regions, i.e., green marked regions.

The process of glaucoma detection through ophthalmologic examinations is mostly done by assessing the increase in IOP. However, glaucoma can occur without this increase. Imaging exams such as optical coherence tomography (OCT), Heidelberg retinal tomography (HRT) and computational methods that evaluate retinal images are ways to detect deficiencies caused by glaucoma. These tests should be reviewed by a trained specialist, which implies a time-consuming and expensive process [4]. Computational methods have enabled more automatic processes of disease detection, resulting in more efficient, safer, and lower-cost diagnosis. Compared to traditional methods, the combination of specialists and computational tools enables a diagnosis with a higher degree of specificity and sensitivity, even in asymptomatic

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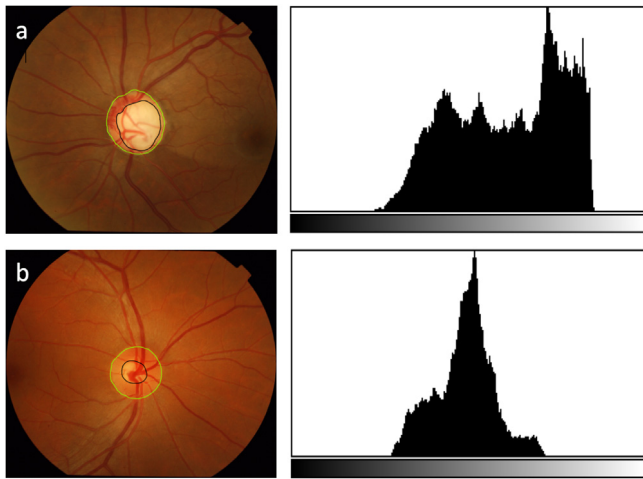


Fig. 1. Drishti-GS image database. (a) glaucomatous image, (b) healthy image.

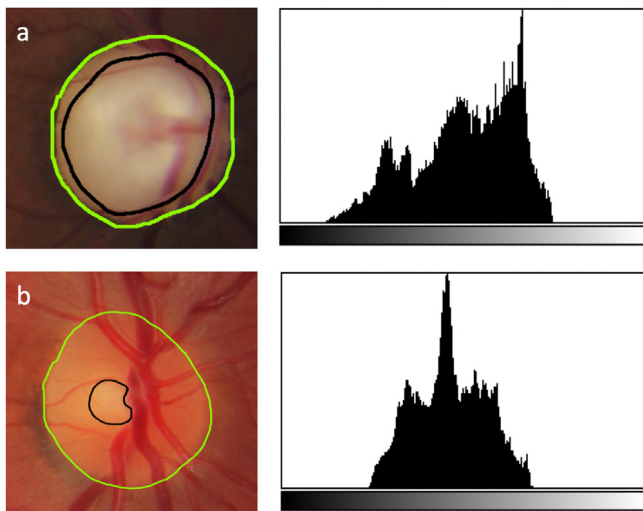


Fig. 2. RIM-ONE image database. (a) glaucomatous image, (b) healthy image.

diseases such as glaucoma [5,6]. The computational methods allow early diagnosis, reducing injuries through treatment in the early stages of the disease.

In this context, this paper aims to develop a computational method that enables the automatic detection and diagnosis of glaucoma, using a generative adversarial network (GAN) and taxonomic diversity index. A GAN was used to delimit the OD region from other structures present in the retinal image. The semantic segmentation resulting from GANs returns ideal markings on different images since pixels are grouped according to the general characteristics of individual objects [7].

GANs can be applied to several computational vision applications, such as object detection [8], high-quality image generation [9,10], semantic segmentation [11], and medical images recognition and classification [12]. Thus, the application of GANs for semantic OD segmentation is justified due to its learning and pattern recognition capabilities. Consequently, this network can discriminate OD markings with adequate training.

The taxonomic indexes were used in the characterization of regions of interest (ROIs) segmented by the GAN. These indexes are based on the species richness and abundance of individuals, that is, on the variability of pixel values and the number of pixels for each value, respectively. Generally, ROIs generated by GANs

are rounded. Given this, we use descriptors that explore the characteristics of ROIs related only to texture. Therefore, this paper presents contributions in three areas: in the health area, offering an automatic methodology to aid in the diagnosis of glaucoma; in computer science, contributing to: (i) the use of GANs for OD segmentation; (ii) the adaptation of techniques from other areas of knowledge to the area of image processing through the proposition of texture descriptors based on taxonomic diversity, and (iii) the proposal of a system with low financial cost and high performance, developed using free technologies; and finally, in the social sphere, providing a low-cost and easy-to-deploy diagnostic methodology that can be used even in places with limited resources and difficult access.

2. Related works

The specialized literature offers studies on the issue of automatic detection and diagnosis of glaucoma, as well as the application of computational techniques that seek this objective. This section highlights some of them. We divided the related works into two groups: (i) works that only use texture or diversity indexes descriptors to characterize images; and, (ii) works that utilize deep learning approaches.

2.1. Texture features and diversity indexes

Yadav et al. [13] propose an automatic classification methodology of glaucoma by applying the gray level co-occurrence matrix (GLCM) and the principal component analysis (PCA). The classification method provides a high level of accuracy, at 90%.

Maheshwari et al. [14] introduce an automatic method for glaucoma diagnosis using private and public image datasets. The first is composed of 60 images, while the second has 500 images from the Medical Images Analysis Group (MIAG). The feature extraction phase uses empirical wavelet transform and correntropic functions. The classification results achieve 100% of sensitivity, 96.67% of specificity and 98.33% of accuracy.

Tuncer et al. [15] present a hybrid methodology for OD segmentation, using L1 minimization and region growing. This work uses a non-public dataset containing 30 retinal images. The proposal achieves promising results, with a Dice score of 0.92 and a Jaccard similarity of 0.87.

Carvalho et al. [1] demonstrate a glaucoma detection method through texture descriptors based on phylogenetic diversity. In this proposal, automatic OD segmentation is achieved through Otsu and k-means algorithms. The authors use the Drishti-GS and RIM-ONE image datasets. The image classification results show 99.6% accuracy.

In de Carvalho Filho et al. [16], a method is proposed for the classification of lung nodules using diversity indexes and genetic algorithms. The results were promising and the authors achieved an accuracy of 92.52%.

The method proposed in Carvalho et al. [17] presents an automatic method for classifying breast lesion images as malignant or benign. The feature extraction is based on phylogenetic indexes. Then, classification is conducted using multiple classifiers. Experiments are performed to verify the performance of the proposed method. The results show that the method achieves 99.73% accuracy, 99.41% sensitivity, and 99.84% specificity.

2.2. Deep learning approaches

Orlando et al. [18] propose an OD and optical cup (OC) detection method that uses two pre-trained convolutional neural networks applied to retinal fundus images. The result using only the Drishti-GS data set achieves the value of 0.762 for the receiver

operating characteristic (ROC), which indicates the method's potential. Also, it provides significant evidence of the importance of preprocessing and data set augmentation for training methods that use Deep Learning.

Sevastopolsky [19] presents an approach for automatic segmentation of the OD and OC using a U-net convolutional neural network modification. The method is tested in the RIM-ONE, DRIONS-DB, and Drishti-GS images datasets. This work presents results of an F-score of 0.95 for segmentation. On the other hand, the OC was more difficult to detect, resulting in an F-score of only 0.85.

Ferreira et al. [7] propose a methodology for automatic detection of glaucoma through retinal images analysis. They use a Convolutional Neural Network (CNN) for segmentation and attributes based on texture to characterize ROIs to support the classification. The proposal evaluation uses the RIM-ONE, DRIONS-DB, and Drishti-GS image datasets. The results exhibit 100% accuracy, showing comparable quality to present OD segmentation methods.

Shankaranarayana et al. [20] propose a Fully Convolutional Network-based architecture and adversarial training for automatic OD and OC segmentation tasks. In this architecture, a RESNET-inspired convolutional network called ResU-net is used, which incorporates residual learning together with a concatenation of resources in an architecture that makes use of a generating and a decoding network. The proposal achieves promising results, with a Dice score of 0.98 and an IOU of 0.96 for OD segmentation.

Jiang et al. [21] present GL-Net, a GAN model based on deep convolutional networks, used for the OD and OC semantic segmentation task. GL-Net consists of two network structures, including a generator and a discriminator. In the construction of the generating network, the deep learning architecture VGG16 was used. The discriminator consists of a convolutional 8-layer network model with the Leaky-ReLU activation function. This paper presents F-score results of 0.97 for OD segmentation using the Drishti-GS image base with a total of 101 sample cases.

In Table 1, we present a survey of the works that have a direct and indirect correlation with the proposed method.

The discussed studies offer promising results for OD segmentation and glaucoma classification, inspiring the methodology proposed in this work. They suggest the need to improve the segmentation of regions and classification techniques. Hence, the present work intends to:

- Employ GANs for OD segmentation in order to achieve better results with a smaller number of images;
- Apply taxonomic diversity indexes for feature extraction and texture patterns discrimination of regions segmented by GANs;
- Utilize extensive image datasets to offer a generic and robust method.

Therefore, we aim to offer an efficient, fast and low-cost glaucoma diagnosis second opinion to specialists.

3. Material and methods

The computational method proposed in this work aims to perform OD detection, feature extraction, and pattern classification to provide a rapid, accurate, and automatic glaucoma diagnosis using retinal images. Thus, the methodology applied is composed of six distinct stages. The first stage consists of data acquisition from the Drishti-GS [22] and RIM-ONE [23] images datasets, following by OD segmentation via a GAN algorithm. The third corresponds to the postprocessing stage, where filling holes and contrast enhancement techniques are applied. The fourth step is the texture features extraction of OD using taxonomic indexes. The fifth stage consists of data classification through the WEKA [24] tool, followed by results validation, the final stage. These steps are illustrated in Fig. 3.

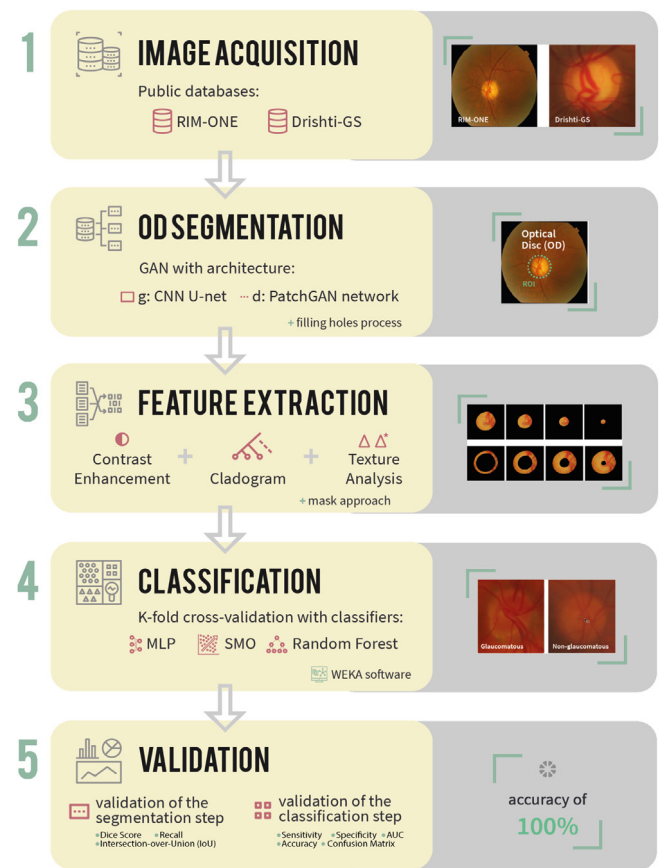


Fig. 3. Proposed method.

3.1. Image acquisition

The Drishti-GS and RIM-ONE v2 are formed by a set of retina images, containing images with healthy and glaucomatous ODs. These datasets gain high acceptance in the scientific community, with several glaucoma analysis and detection applications.

The Drishti-GS dataset has 101 eye images, of which 70 are glaucomatous, and 31 are healthy. All images were collected by clinical researchers at the Aravind Eye Hospital, in India. The patients were selected with an age range from 40 to 80 years and equally divided by gender. Each image has manual markings of OD and OC made by four experts [22].

The Canaria University Hospital, the San Carlos Clinic, and the Miguel Servet University Hospital released the RIM-ONE v2 dataset, which has 455 retinal images analyzed by specialists. Among them, 200 are images with glaucoma at several levels, and 255 are healthy images. Each image has a corresponding mask [23]. Fig. 4(a) and (b) present samples of images contained in the datasets described above.

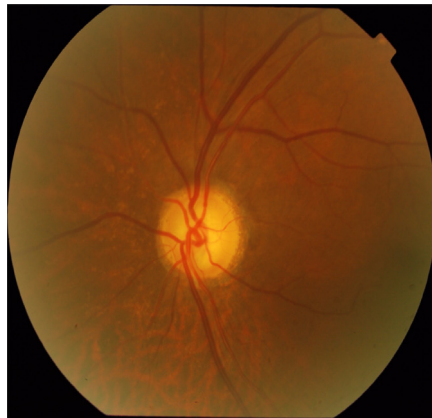
3.2. OD segmentation

In the present methodology, we apply the GAN as a tool for semantic OD segmentation. The GAN seeks to solve problems with a high level of abstraction, based on procedures of estimation by generative models. These models require only the specification of a highly abstracted objective so that it can learn automatically, altering its loss function appropriately to meet the stated goal. Thus, the GAN can be applied to a large group of tasks that traditionally required very different types of network and different types of loss functions [25–27].

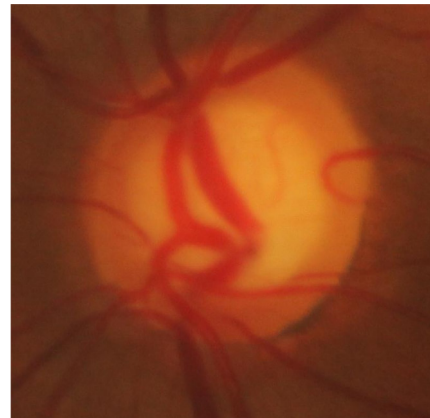
Table 1

Summary of related work and correlation with the proposed method.

Work	Techniques	Dataset	Sample	Evaluation metrics	Goal/Correlation
[13]	GLCM and PCA	Private dataset	20	Accuracy(%) = 90	Automatic classification of glaucoma based on feature extraction in images
[14]	Empirical wavelet transform and correntropic functions	Public dataset MIAG	560	Accuracy(%) = 98.3; Specificity(%) = 96.6; Sensitivity(%) = 100	Automatic glaucoma classification method based on feature extraction in images
[15]	L1 minimization and region growing	Private dataset	20	Dice score = 0.92	Automatic optical disc segmentation
[1]	Otsu, k-means and phylogenetic diversity	RIM-ONE and Drishti-GS	556	Dice score = 0.81; Accuracy(%) = 99.6; Specificity = 99.3; Sensitivity(%) = 100; AUC = 0.9	Automatic optical disc segmentation and automatic glaucoma classification based on feature extraction in images
[16]	Diversity indexes and genetic algorithms		1405	Accuracy(%) = 92.52	Automatic classification of lung nodule using diversity indexes
[17]	Phylogenetic indexes	DDSM database	1155	Accuracy(%) = 99.73; Specificity(%) = 99.84; Sensitivity(%) = 99.41	Automatic breast cancer classification using texture features
[18]	Pre-trained convolutional neural network	Drishti-GS, DRIVE	–	AUC = 0.7	Automatic glaucoma classification methodology
[19]	U-net convolutional neural network	RIM-ONE v1, DRIONS-DB and Drishti-GS	370	Dice score = 0.95; IOU = 0.89	Automatic optical disc segmentation
[7]	Convolutional neural network	RIM-ONE v1, DRIONS-DB and Drishti-GS	370	Dice score = 0.84; Accuracy(%) = 100; AUC = 1	Automatic optical disc segmentation and automatic glaucoma classification
[20]	Fully convolutional and adversarial networks	RIM-ONE v1	159	Dice score = 0.98; IOU = 0.96	Automatic optical disc segmentation using GANs
[21]	Deep convolutional generative adversarial networks	Drishti-GS	101	Dice score = 0.97	Automatic optical disc segmentation using GANs
Our method	Conditional generative adversarial networks and taxonomic diversity indexes	Drishti-GS RIM-ONE	556	Dice score = 0.89; Accuracy(%) = 100; AUC = 1	Automatic optical disc segmentation and automatic glaucoma classification



(a)



(b)

Fig. 4. (a) Healthy retinal image4 from the Drishti-GS dataset and (b) healthy retinal image from the RIM-ONE v2 dataset.

The success achieved by the generative adversary models is mainly justified by the way their architectures are arranged, and how their training improves the results [28]. GANs are generally composed of two competing neural networks, called generators and discriminators. The generating network receives input and generates samples. The discriminating network receives the generated samples and the training data to distinguish between the two types of information. Thus, the generator will learn to produce more realistic samples, and the discriminator, in parallel, will learn to distinguish the generated data from the actual data, consolidating the model learning [25,29].

We extended the GAN structure to the conditional configuration (cGAN). There are no restrictions on the data modes generated by the original GAN configuration. However, if we

provide additional information to the generating network, the GAN will generate images with the desired characteristics [30]. The cGAN adopts class labels to produce conditional samples. These samples are provided to the generator and discriminator, significantly improving the quality of the samples [31]. Fig. 5 shows the architecture of the adversarial training adopted. Thus, cGAN is suitable for the OD segmentation task, where a retinal image is conditioned as input, and an OD marking image is generated as a corresponding output.

We adapted the generator and discriminator architectures presented in [26] to segment the retinal image and to separate the ROI from the OD of the other structures present in the image. The adapted architecture consists of a cGAN named *Pix2pix*, based on image-to-image translation, which uses modules of the

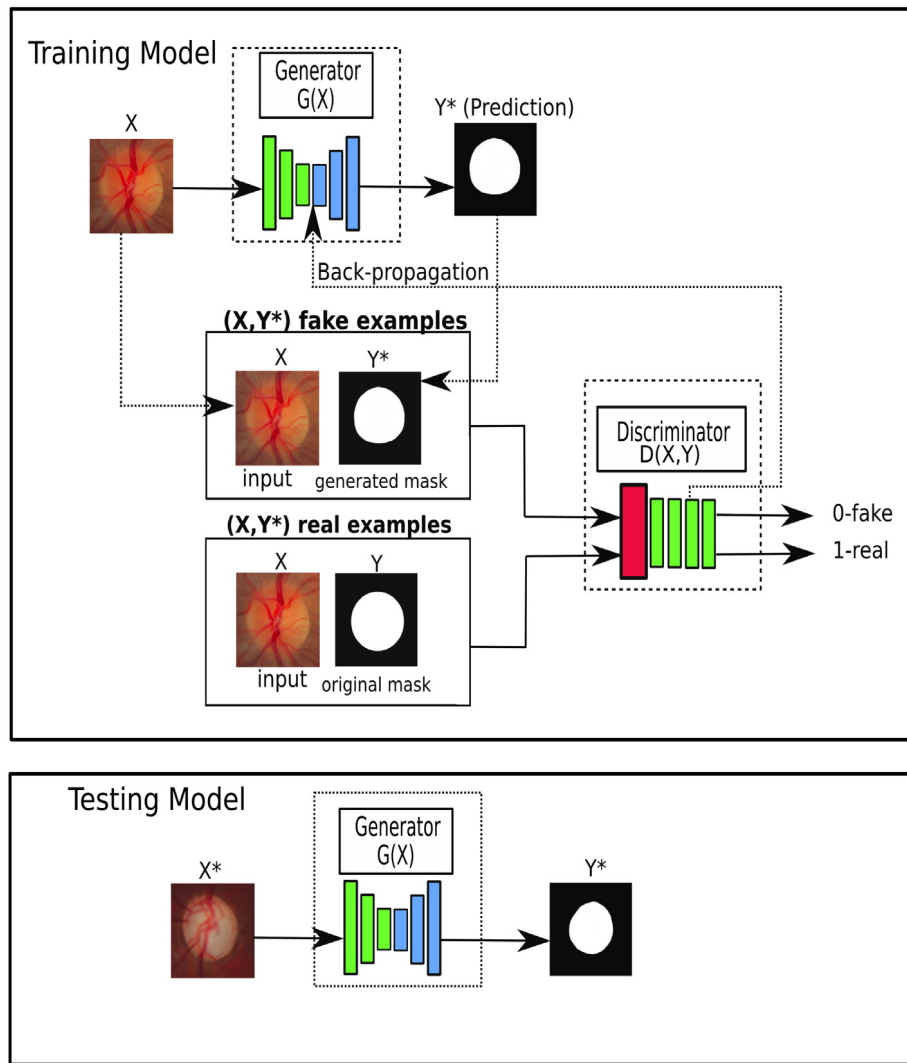


Fig. 5. A GAN scheme with one generating network G and one discriminating network D .

convolution-BatchNorm-ReLU [32] form applied to the generator and the discriminator.

The CNN *U-net* [33] was used as the generator. This CNN is composed of two paths: a contraction path (downsampling) and an expansion path (upsampling). The first one followed the typical convolutional network architecture, except the fully connected layers. We applied the 4×4 convolution to build the contraction path, each convolution followed by a LeakyReLU activation function and a BatchNormalization operation. In this way, we have the activation functions from the previous layer normalized in a batch, with the mean close to 0 and the standard deviation close to 1. This operation improves the network speed, performance, and stability, making it easier to generalize and avoiding the use of dropout to reduce over-tuning [32].

The expansion path used resource mapping and convolutions to map the corresponding resources. Each step of the expansion path consists of a mapping of the characteristics (deconvolution) followed by a 4×4 convolution having an activation ReLU function and a BatchNormalization operation. In this way, the network can recognize patterns at various scales [7].

We added *skip connections* between each layer i and layer $n - i$, so all channels in layer i are concatenated with those in layer $n - i$, where n is the total number of layers. This change enables low-level information to be shared between the downsampling and

upsampling layers, transferring this information directly over the network. Fig. 6 illustrates such architecture.

Finally, the PatchGAN network was used as a discriminator of the *Pix2pix* model. PatchGAN is a conventional convnet [34], composed of a concatenation layer and convolutional encoders, similar to the downsampling layers of the generator, as shown in Fig. 7. The network architecture used differs from a conventional neural network by the convolutional and pooling layers, which efficiently extract features of an image using fewer dimensions. Thus, this discriminator architecture is capable of mapping an image to an $N \times N$ array of X outputs, where each X_{ij} means a patch ij in the image, classifying it as real or false. This discriminator effectively models the image as a Markov random field [35], assuming independence between pixels separated by more than one pixel of diameter from the patch. The literature indicates that PatchGAN is an appropriate choice to evaluate the cGAN and to reinforce its training [26].

Table 2 presents a summary with the other configuration parameters used in the proposed architecture.

We chose to use the cGAN for the following reasons:

- As presented in [36], a CNN achieves good generalization with large data sets;
- The cGAN has the advantage that the generator network, is not updated directly from training images, but with gradients provided by the discriminator network facilitating

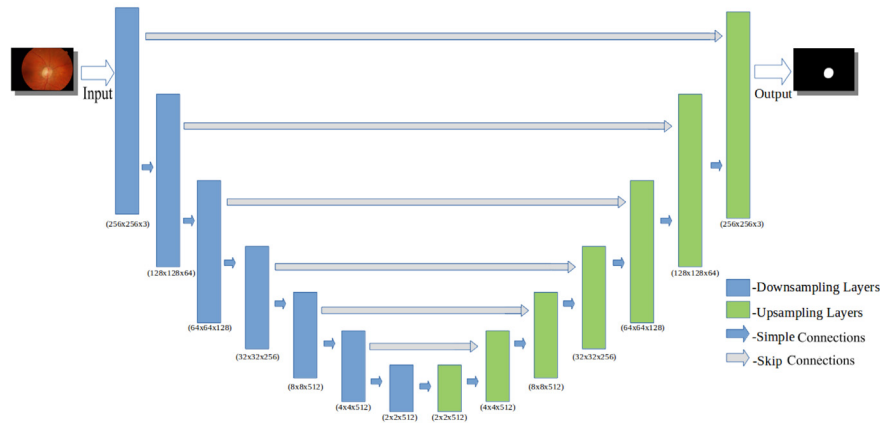


Fig. 6. U-net architecture.

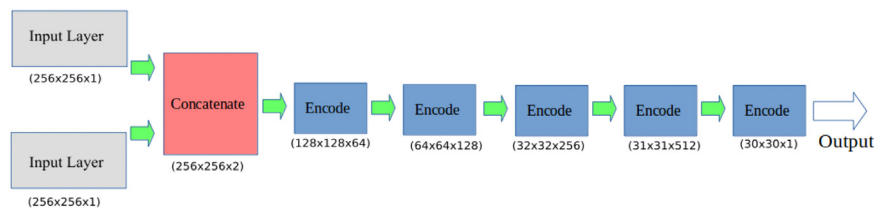


Fig. 7. PatchGAN architecture.

Table 2

Hyper-parameters used for cGAN training.

Network	Loss functions	Optimizer	Batch-size	Learning rate
Generator network (Unet)	Mean Square Error (MSE)	Adam	1	0.0002
Discriminator network (PatchGAN)	Mean Square Error (MSE) and Mean Absolute Error(MAE)	Adam	1	0.0002

learning and making the opposing model more suitable for small sample datasets than the CNN [25].

Based on this, we chose to use the best of each approach, believing that they are sufficient to circumvent problems related to (i) not enough images for the training, validating and testing of the method; and (ii) proposing a new approach to diversify studies to provide more promising results. Thus, we propose a robust architecture, implemented in low-cost hardware, capable of providing results closer to the expert's markings, thus enabling an analysis of image regions capable of providing important image properties.

3.2.1. Training

Once the architectures were defined, the next step was to select the input data for network training. The entire set of images was divided into training/validation and test sets: 80% for training and 20% for tests. The training set consists of 445 randomly selected images, of which 364 are from RIM-ONE and 81 from the Drishti-GS dataset. However, even using more than one dataset, the number of images available to train the model is small, as demonstrated by the examples analyzed in the bibliography [7, 18].

Considering that the selected datasets present a small number of samples, we used two techniques as an alternative for dataset augmentation. Such techniques were applied only to the training images. The first technique consists of rotating the images of the retina and their masks by 10 degrees. This process is repeated from 0 to 350 degrees, generating 35 new samples.

The second data augmentation technique consists of re-scaling the retinal images and their respective masks, performing changes in the original image, both decreasing and increasing the proportions. We re-scale from 5% to 30% of the image in steps of 5%, resulting in 12 new images. Finally, the training dataset has a total of 21 360 images, among them 445 originals, 15 575 rotated and 5340 re-scaled. After that, we converted all these images to grayscale to facilitate the training via analysis of only one channel. Fig. 8 and Fig. 9 show examples of the training images resulting from the data augmentation process.

After the data augmentation procedures, we defined the GAN input parameters, which are crucial to determining the objects that the GAN must learn to recognize. In our methodology, the retinal images were the input of the generating network training. The generated output and the masks of input images were simultaneously used to train the discriminator (Fig. 5). In this way, each mask represents the target for the network. Thereby, we expected the GAN to recognize the OD properties to produce an adequate segmentation of this region.

We execute tests during the training to verify the adequate number of epochs (training cycles) that provides an acceptable image of the segmentation. These tests helped to analyze the learning ability of the model and resulted in 100 epochs for a satisfactory segmentation. A higher number of epochs will imply the same result. Fig. 10 shows the training results with various numbers of epochs.

As shown in Fig. 10, some noises appear in the original GAN segmentation. Fig. 11 shows the loss model over the number of epochs, which indicates that from 80 epochs the loss stabilizes

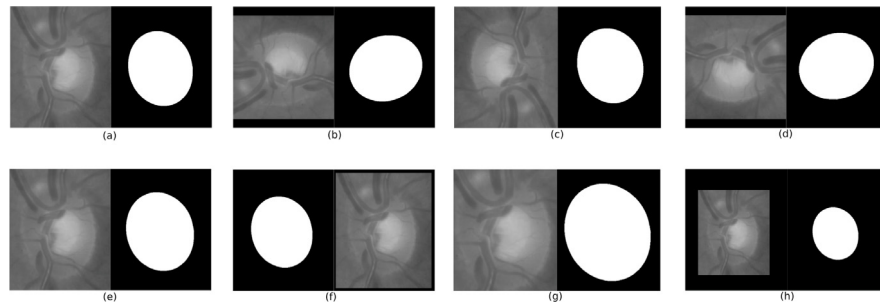


Fig. 8. Example of rotated images with scale variations used in the cGAN training in RIM-ONE image database: (a) original image, (b) image rotated 90 degrees, (c) image rotated 180 degrees, (d) image rotated 270 degrees, (e) image scale increased by 5%, (f) image scale decreased by 5%, (g) image scale increased by 30%, and (h) image scale decreased by 30%.

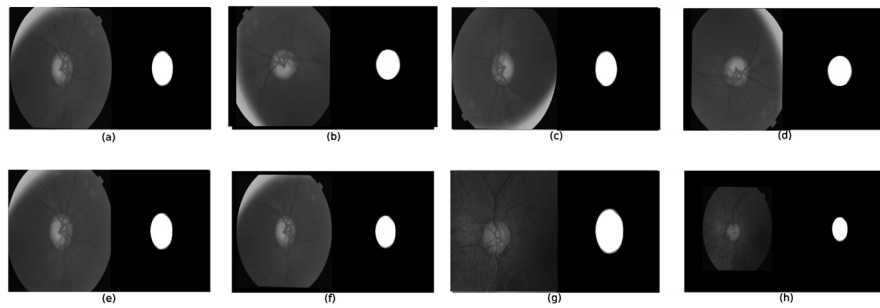


Fig. 9. Example of rotated images with scale variations used in the cGAN training in Drishti-GS image database: (a) original image, (b) image rotated 90 degrees, (c) image rotated 180 degrees, (d) image rotated 270 degrees, (e) image scale increased by 5%, (f) image scale decreased by 5%, (g) image scale increased by 30%, and (h) image scale decreased by 30%.



Fig. 10. (a) Original image, (b) ground truth, (c) segmentation with 30 epochs, (d) segmentation with 50 epochs, (e) segmentation with 100 epochs, and (f) segmentation with 150 epochs.

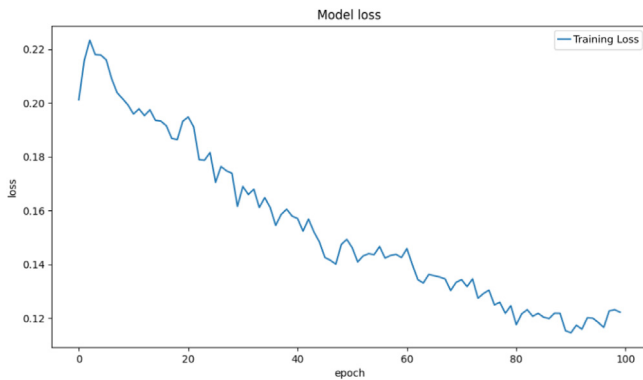


Fig. 11. Loss model for training with 100 epochs.

and the algorithm no longer learns. Due to the hardware limitations in our testing environment (64-bit Core I5 processor and 4 GB RAM), the tests were performed with 100 epochs, and they take more than one week to complete the training. Free deep learning environments such as Google Colab and Kaggle limit the GPU processing by either the execution time or the number of images.

Thus, we chose to use the images resulting from the experiment with 100 epochs due to the small variation of results

when compared to the results with a higher number of epochs (Fig. 10(f)). The noise removal approach applied is based on filling holes [37]. Fig. 12 demonstrates the segmentation results after noise removal.

In addition to the presented information, we developed all the algorithms related to the methodology in the Python 3.5 programming language with the Keras framework [38] and the backend TensorFlow [39]. For images manipulation, the OpenCV library [40] was adopted.

3.3. Feature extraction

An image feature extraction aims to determine aspects that result in some quantitative information or that can be useful for discrimination between classes. As highlighted by Pedrini and Schwarcz [41], the complexity of image analysis lies in the definition of a set of features that can describe each region contained in the image.

Although deep learning demonstrates promising results to represent information in computer vision, such supervised models require large amounts of labeled data and computing resources, which implies a high cost [36]. In this scenario, this work presents a traditional methodology for feature extraction based on texture exploration. Such methodology was applied to a small dataset to demonstrate its applicability for the characterization of glaucoma. In such methodology, taxonomic diversity

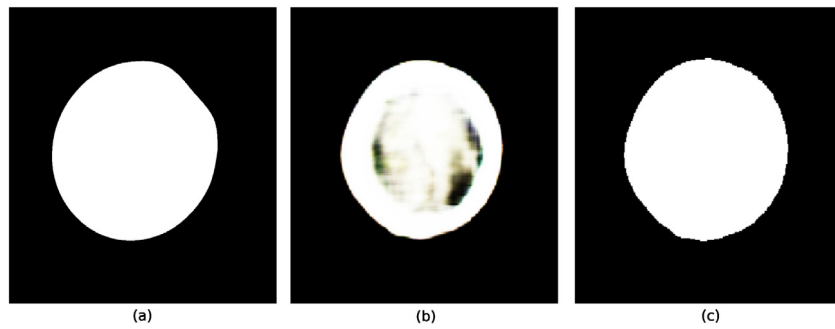


Fig. 12. (a) Ground truth, (b) GAN segmentation, (c) filling holes for noise removal.

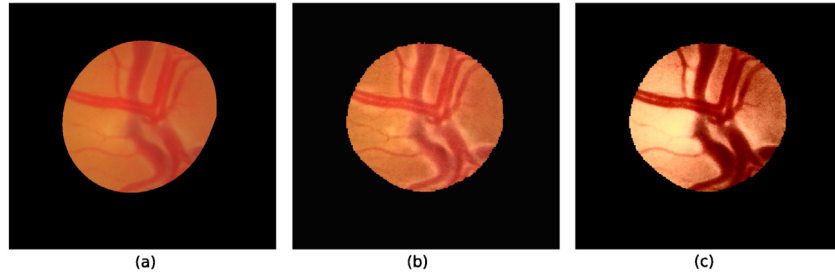


Fig. 13. (a) Original image, (b) histograms equalization, and (c) CLAHE.

and taxonomic distinction indexes were used as descriptors. In the subsequent sections, we describe the following items:

- Contrast enhancement techniques applied in ROIs segmented by the cGAN;
- The fundamentals of organization of the phylogenetic tree;
- The texture descriptors used in our method based on distinction and taxonomic diversity.

3.3.1. Contrast enhancement

Contrast equalization was used in our methodology to map the values of luminous intensity to similar bands between the analyzed images. Thus, the variations of contrast between the images become similar, avoiding the influence of peaks of intensity in the feature extraction results.

We apply the histogram equalization and the contrast-limited adaptive histogram equalization (CLAHE) techniques for contrast enhancement. Histogram equalization is a simple method for image enhancement that maps the image so that the resulting histogram converges to the uniform distribution [42]. Fig. 13(b) presents the result of the histogram equalization technique.

CLAHE merges the histogram equalization approach with a contrast limitation. Thus, the histogram equalization is modified by enhancing the contrast, limited to a pre-defined value [43]. In this work, we set the contrast limit to 40, which is a standard value of the OpenCV library. We tested different values, but the standard one presented the best result. Hence, parts of the histogram above the specified limit are cut out and uniformly distributed.

Finally, histogram equalization [44] was applied, and Fig. 13(c) demonstrates the result.

3.3.2. Phylogenetic tree: Cladogram

The taxonomic diversity is based on the set of distances between pairs of species accumulated from the taxonomic trees [45]. In the context of phylogenetic relationships, taxonomic trees or simply phylogeny are represented by a tree where the leaves represent the organisms or species, and the inner nodes represent

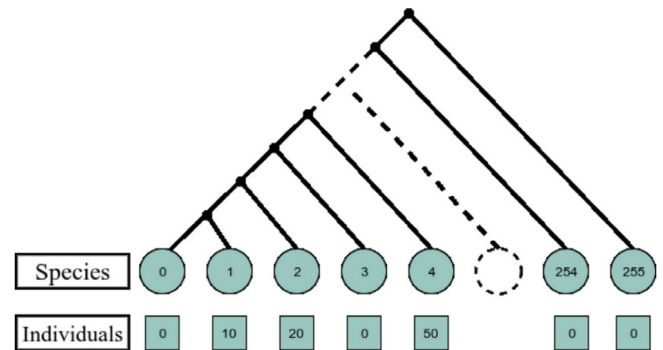


Fig. 14. Rooted tree in the form of a sloped cladogram [7].

the ancestors. The edges of the tree denote the evolutionary relations [46].

The cladogram is a representation of the phylogenetic tree. In this work, we use the sloped cladogram [46], as shown in Fig. 14. These trees allow us to apply the indexes that connect diversity, and parenthood between species. Thus, the information used to calculate the taxonomic diversity indexes can be summarized by a matrix of distances between species, which is obtained from the cladogram.

3.3.3. Taxonomic diversity indexes

Taxonomic diversity is very sensitive to environmental disturbances, and it is appropriate to evaluate the differences between communities [47]. A community with several genera must present a greater diversity than a community where the species belong to the same genus. Therefore, the analysis of taxonomic diversity provides the understanding of a particular community pattern [45].

The application of diversity indexes to image processing results in the discrimination of characteristics in an image, where, in the context of image processing, we have the following:

1. The image or an image region represents the *community*;

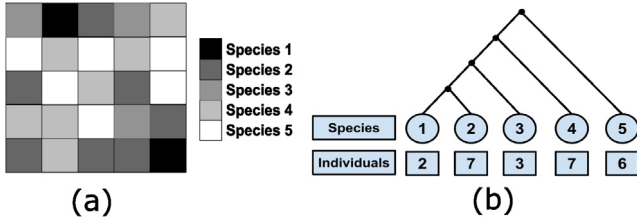


Fig. 15. (a) Representation of the analyzed region, (b) extracted phylogenetic tree.

2. The values of the pixels represent the *species*; and,
3. The *individuals* correspond to the number of pixels belonging to the same species.

The Taxonomic Diversity (Δ) and Taxonomic Distinction (Δ^*) indexes represent the calculation between two organisms chosen randomly from a phylogeny inside a community [48]. As stated by de Carvalho Filho et al. [49], these indices have three essential factors for application: the number of species, number of individuals, and the structure of connection between species (number of edges). In this work, these two indices are used as texture descriptors in glaucomatous and healthy retinas images.

The value of Δ considers the abundance of the species and the taxonomic relation between them. Therefore, its value expresses the average taxonomic distance between two individuals, chosen randomly from a sample [16]. This is defined by Eq. (1), where X_i is the abundance of the i th species, N is the total number of individuals and ω_{ij} is the distance of the species i to the species j in the taxonomic classification.

$$\Delta = \frac{\sum \sum_{i < j} \omega_{ij} X_i X_j}{[N(N-1)/2]} \quad (1)$$

The Taxonomic Distinction Index (Δ^*) represents the average taxonomic distance between two individuals, with the restriction that they are from different species [16]. This index is defined by Eq. (2), where $X_i (i = 0, \dots, s)$ is the abundance of species i , $X_j (j = 0, \dots, s)$ mean the abundance of species j and ω_{ij} is the distance of the species i to the species j in the taxonomic classification.

$$\Delta^* = \frac{\sum \sum_{i < j} \omega_{ij} X_i X_j}{\sum \sum_{i < j} X_i X_j} \quad (2)$$

For each ROI analyzed, we extracted two taxonomic diversity indexes. As our methodology differential, we calculated the taxonomic diversity and taxonomic distinction for the pixels in gray levels, as in [49], and for each pixel of the images generated from the RGB channels. In the end, the extracted attributes seek to provide measures of homogeneity or heterogeneity for each ROI.

To exemplify the feature extraction process through indexes, we created Fig. 15.

Fig. 15(a) is an example of an image with five species and 15(b) presents the generated cladogram with their respective species and individuals. The information on species and individuals represented in the cladogram is fundamental for the application of the indexes. Therefore, we have information about kinship, richness, and abundance of species. From this information, we calculate and present the results for each proposed index: $\Delta = 2.92$, and $\Delta^* = 2.33$.

Finally, we present Fig. 16 which shows the actual result for OD image calculations.

3.3.4. Internal and external masks approach

This approach aims to uncover patterns of diversity in areas near the edge and the inner areas of the ROI. The approach

Table 3

Hyper-parameters used for classifier training.

Classifier	Hyper-parameters
MLP	Neurons in hidden layer = 57; Learning rate = 0.3; Momentum = 0.2; Training time = 500
SMO	Complexity parameter = 1.0; Epsilon = 1.0E12; Kernel = PolyKernel; Seed = 1; Tolerance Parameter = 0.001
RF	Number of trees = 100; Seed = 1

with internal masks consists of reducing the scale of the original masks. The binarization of the original ROI was applied to create the first mask. The second mask is created based on the decrease of the scale in relation to the first one by the center of mass. The following masks were acquired by originating from their previous ones in a sequence from the more external to the more internal. The scales applied in this study were: 20%, 40%, 60%, and 80% [50]. Fig. 17 shows examples of the generated masks.

At the end of the process, we have a set of 8 masks (4 internal and 4 external) for each image.

3.4. Classification

For pattern analysis, we have chosen some classifiers taking into account their wide adoption in related literature, their adaptability to our experiments, and their efficacy. Among the analyzed classifiers, three were highlighted and selected for classification stage: Multilayer perceptron (MLP) [51], Sequential Minimal Optimization (SMO) [52], and Random Forest [53]. The classifiers are evaluated using the k-fold cross-validation method [54].

The classification process was performed using the WEKA software, which aggregates an extensive set of machine learning algorithms and allows the development of new approaches [24].

The parameters used in the classifiers were the defaults of the WEKA tool. Table 3 presents the summary with the configuration parameters used.

3.5. Validation

After completing the classification stage, it is necessary to validate the results and discuss possible improvements. This methodology uses metrics that are commonly applied to image processing systems. The validation occurred in two steps: first, we validated the segmentation phase; then, we validated the classification phase. In the evaluation of segmentation, the following metrics were adopted: Dice score, Intersection-over-union (IOU), and Recall.

The Dice Score is calculated by Eq. (3) and allows us to verify the result of the segmentation (A) about the original mark made by a specialist (B) [19].

$$\text{DiceScore} = \frac{2|A \cap B|}{|A| + |B|} \quad (3)$$

IOU (Eq. (4)) is an evaluation metric commonly used to evaluate semantic segmentation that considers the similarity between the binary masks generated by our method prediction and the specialist marks [55].

$$\text{IOU} = \frac{|A \cap B|}{|A \cup B|} \quad (4)$$

On the other hand, the Recall (Eq. (5)) metric shows how well the segmentation method detects the region belonging to the specialist mark.

$$\text{Recall} = \frac{|A \cap B|}{|A \cup B| + |A - B|} \quad (5)$$

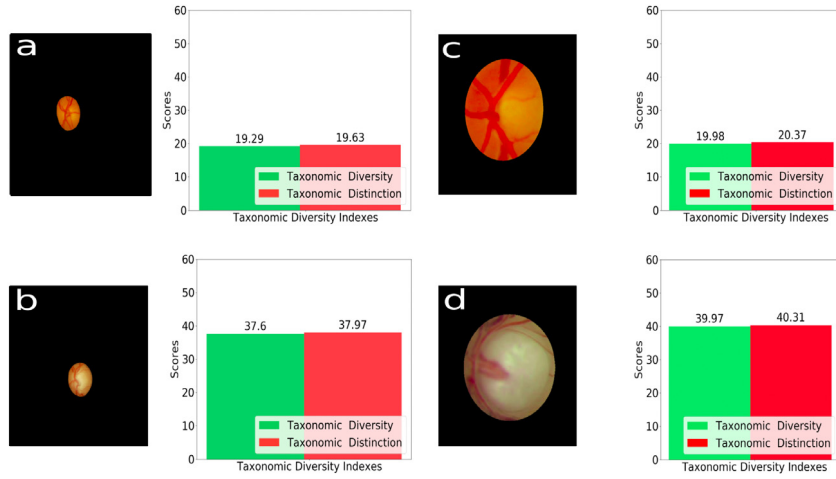


Fig. 16. Result of the indexes extracted in red channel with real examples: (a) healthy image in Drishti-GS image database, (b) glaucomatous image in Drishti-GS image database, (c) healthy image in RIM-ONE image database, and (d) glaucomatous image in RIM-ONE image database.

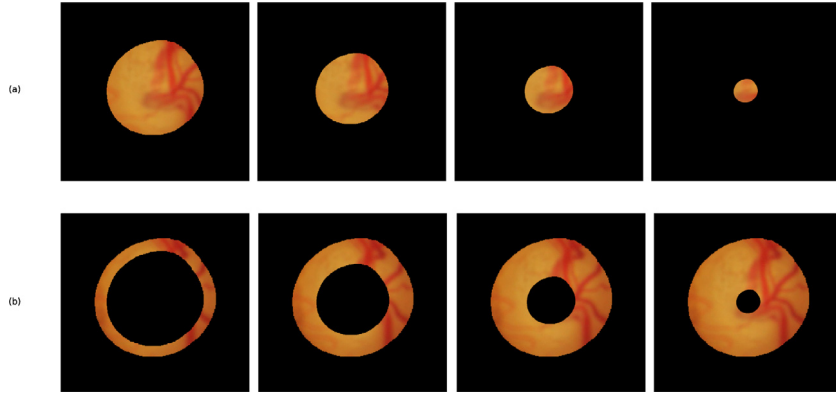


Fig. 17. (a) Internal masks and (b) external masks.

In order to evaluate the performance of classification models, the following statistics are generally used: Sensitivity (Sen), Specificity (Spe) and Accuracy (Acc). These metrics make use of the confusion matrix, which indicates the correct or incorrect classification of the classes in use, grouping the results into four classes: False Negative (FN), False Positive (FP), True Positive (TP), and True Negative (TN).

Sensitivity (Sen) is the ability of a diagnostic test to identify true positives in individuals with the disease. The function to indicate the effectiveness of the method is defined according to (Eq. (6)).

$$Sen = \frac{TP}{TP + FN} \quad (6)$$

Specificity (Spe) is the ability of a diagnostic test to identify true negatives in truly healthy individuals (Eq. (7)).

$$Spe = \frac{TN}{TN + FP} \quad (7)$$

Accuracy (Acc) calculates the ratio of success, that is, the total of truly positive and truly negatives of the sample, as Eq. (8) demonstrates.

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

In addition to the metrics mentioned earlier, a widely used method for quantitative performance evaluation is the Receiver Operating Characteristic (ROC) [56] curve analysis. As seen in Van Erkel and Peter [56], the area under the ROC curve (AUC) is a

valuable way to evaluate a binary classification system, because it implies the ability of the diagnostic system to discriminate individuals from a sample, taking into account the values of Sen and Spe of the test. The best performance is achieved when the AUC value approaches 1 (equivalent to 100%).

4. Results and discussions

In this section, we present the results obtained by the proposed methodology, including the segmentation with the cGAN and the classification using the taxonomic diversity indexes descriptor. Then, we compare our results with those from related works and provide some discussion.

Initially, to evaluate the methodology, we use separate images as shown in Section 3.2.1. The test set corresponds to 20% of the original image base, containing normal and glaucomatous images. It has a total of 111 images, of which 92 are from the RIM-ONE v2 base and 19 from the Drishti-GS base.

4.1. Results of GAN segmentation

The GAN was applied in the test set composed of 111 retina images. Fig. 18 shows the results from cGAN segmentation as follows: Fig. 18(a) and (c) show examples of RIM-ONE and Drishti-GS images, respectively, and Fig. 18(b) and (d) show the segmentation results.

In order to assess the OD segmentation step, we use a comparison of the ROIs segmented by the cGAN and markings made by

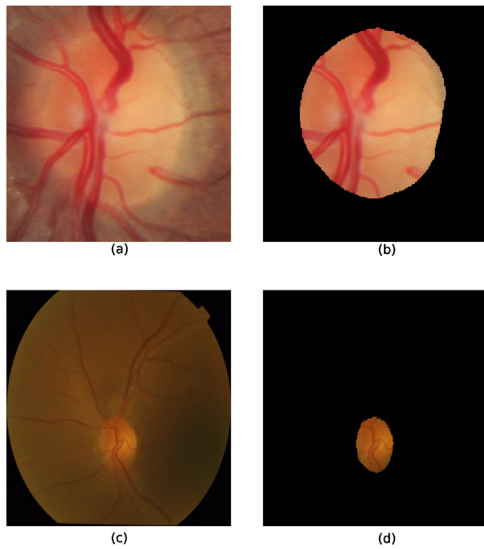


Fig. 18. (a) and (c) Original images from RIM-ONE V2 and Drishti-GS datasets, (b) and (d) ODs after cGAN segmentation.

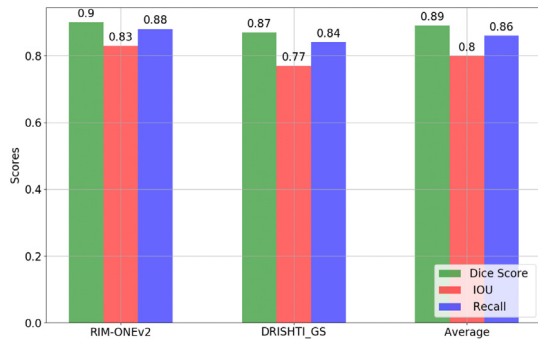


Fig. 19. Evaluation of cGAN OD segmentation.

specialists. The metrics used for evaluation were: IOU, Dice score and Recall. The results are shown in Fig. 19.

As shown in Fig. 19, the segmentation of the RIM-ONE database yielded results that were better than the Drishti-GS database. This is because RIM-ONE has a more substantial amount of images for training. However, even in this context, the Drishti-GS database presented significant results for segmentation, justified by its small variation in the location of the OD region, thus allowing robust learning, even with inferior results. The slight variation in the location from the OD region to the Drishti-GS base is because all images were taken centered on the OD with a field-of-view (FOV) of 30 degrees and dimensions of 2896×1944 pixels [22]. The cGAN OD segmentation goal was satisfactorily achieved, reaching an average Dice score of 0.89.

We chose to use the cGAN only in the segmentation stage because of the promising results available in the literature for this type of problem. Also, another critical factor is the limited number of images available for training, which is another key reason for using cGANs [11,26,31,57–60]. We intend to provide an accurate method for an efficient and low-cost diagnostic for remote and needy regions in the world. Next, we will present a successful and a failing case study for the segmentation stage.

Fig. 20(a) and (b) show the successful cases in OD segmentation, where the green contour represents the marking made by the specialist and the blue contour represents the output obtained by the proposed method. Fig. 20(c) shows the segmentation result for the images of RIM-ONE v2 and Drishti-GS, when using these

two bases for training. For the best case of the RIM-ONE v2 base, we obtained the following evaluation metrics: Dice 94%, IOU 97% and Recall 97%; for the success case of the Drishti-GS base: Dice 91%, IOU 95% and Recall 93%. Our methodology presents a segmentation that is close to the marking made by the specialist, which evidences a satisfactory result for the segmentation. As indicated before, the method presents some difficulty in fitting perfectly to the edges of the OD because this region does not have well-defined edges, making it difficult to achieve a precise segmentation.

Fig. 20(c) and (d) present the cases where the proposed method achieved poorer results for OD segmentation. The green contour represents the mark made by the specialist, and the blue contour represents the result obtained by the proposed method. The evaluation metrics (see Fig. 20(c)) present the worst result among the tests for the base RIM-ONE v2: Dice of 18%, IOU of 30% and Recall of 18%. The worst case for the Drishti-GS base segmentation resulted in a Dice of 64%, an IOU of 71% and a Recall of 65%. The failure cases demonstrate that correctly identifying the OD could be challenging for cases in which the images present small differences between the eye background region and the OD region. In addition, we highlight that the region delimited by the specialist in Fig. 20(c) is smaller than the one which corresponds to the OD region.

Section 4.2 exhibits the results of ROIs classification using the descriptors proposed in our method (Section 3.3).

4.2. Results of classification with taxonomic diversity indexes

After applying the cGAN in the test set, we submitted the resulting ROIs to the characterization process using the proposed texture descriptors. The evaluation of the classifiers was done through k -fold cross-validation, where $k = 10$ (Section 3.4), and the metrics of evaluation described in Section 3.5. The tests using the diversity indexes (Δ) and taxonomic distinction (Δ^*) were applied in each RGB channel and the grayscale image, totaling two features per image, i.e., for each proposed index a feature is computed. Figs. 21 and 22, present the results for the evaluation metrics (Section 3.5) applied in this work.

In the tests with separate channels (Figs. 21 and 22), we obtained significant results, reaching an accuracy rate of 93% for the Random Forest classifier with the red channel (Fig. 22(b)). These results indicated the efficiency of the features extraction methodology and motivated us to perform tests using all the channels together.

The tests using all channels (Fig. 23(a)) showed significant improvement, as a success rate of more than 96% using the MLP classifier was achieved. Besides, these test results showed a good balance among the three evaluation metrics adopted in this work. For this test, eight features were extracted, two for each image of the RGB channels and two of the image in gray levels.

The results presented in Fig. 23(b), correspond to the tests performed using the internal and external masking approach (see Section 3.3.4). For each RGB channel and grayscale image, eight new images corresponding to the masks are generated, totaling 32 images (4×8). Thus, we have four original images (RGB channels and grayscale) plus 32 masks, totaling 36 images. After applying the two indexes in the 36 images, we have a total of 72 features (2×36). Thereby, the results presented a balance improvement of metrics and positive results for all selected classifiers. This improvement is due to the possibility of analyzing OD regions separately.

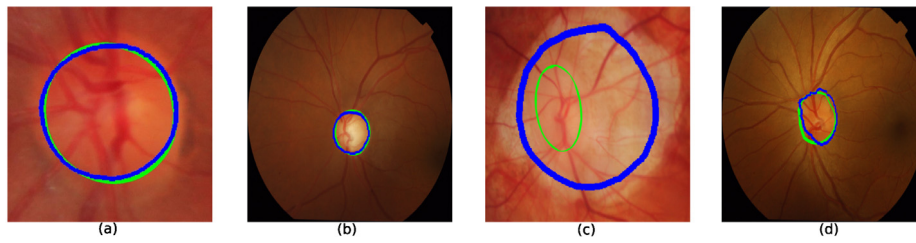


Fig. 20. (a) Best result obtained for OD segmentation in the RIM-ONE v2, (b) best result obtained for OD segmentation in the Drishti-GS, (c) worst result obtained for OD segmentation in the base RIM-ONE v2 and (d) worst result obtained for OD segmentation in the Drishti-GS base.

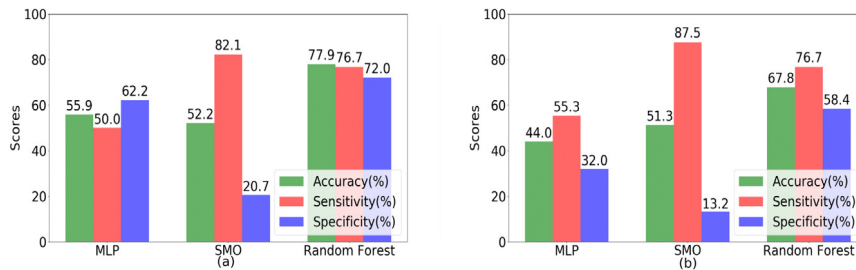


Fig. 21. (a) Tests with grayscale and (b) tests with the blue channel.

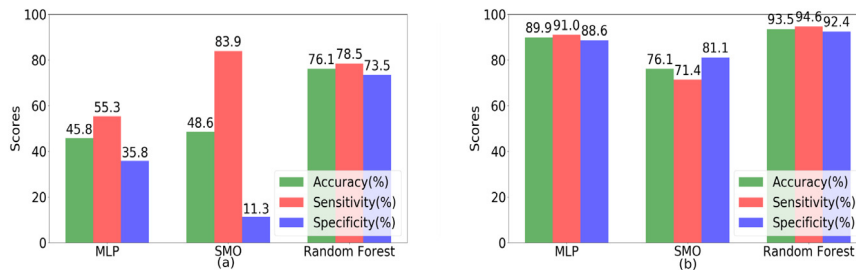


Fig. 22. (a) Tests with the green channel and (b) tests with the red channel.

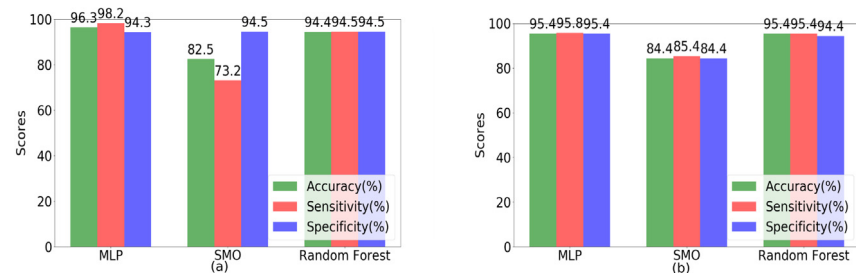


Fig. 23. (a) Tests with all channels and (b) tests with all channels, grayscale and internal and external masks.

4.3. Classification results with contrast enhancement postprocessing

After examining each case, we noticed that the incorrectly classified images had a very similar contrast to the opposite class. Fig. 24 presents an image without the presence of glaucoma, erroneously classified as glaucomatous due to its contrast. Thus, we apply contrast enhancement techniques (see Section 3.3.1) in all images of the test set to highlight properties related to each class of images.

After applying the techniques for contrast enhancement (see Fig. 25), we evaluated all the descriptors together, since they offer the best result combination, as illustrated in Fig. 23(a). The results improvement is explicit for all classifiers, and in the tests with histograms equalization, two of the classifiers were able to correctly identify all the images, reaching an accuracy of 100%.

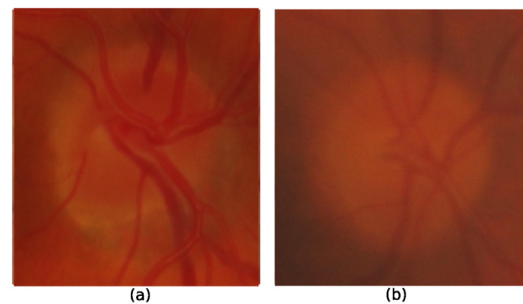


Fig. 24. (a) Example of an image correctly classified as glaucomatous and (b) example of an image incorrectly classified as glaucomatous.

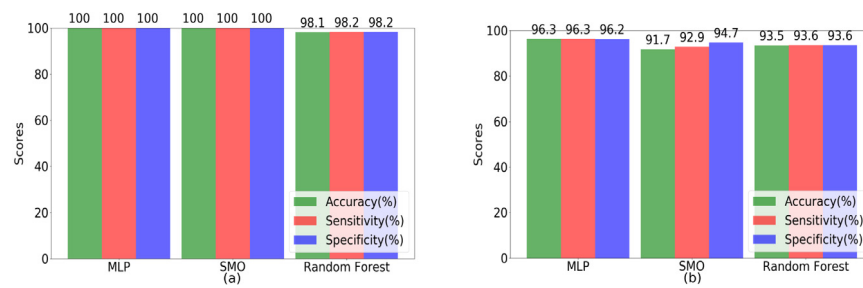


Fig. 25. (a) Tests with all channels and histograms equalization and (b) tests with all channels and CLAHE.

Table 4

Tests with GLCM descriptor, for distance 1.

Classificador	Acc (%)	Spc (%)	Sen (%)	AUC
MLP	67.8	32.3	67.9	0.72
SMO	69.7	30.7	69.7	0.69
Random Forest	69.7	30.1	69.7	0.77

Finally, the application of enhancement techniques with mask approaches achieved the best results. Fig. 26(a) reveals that the results were maintained for all classifiers. However, the results with CLAHE application and the mask approach obtained an accuracy of 100% (Fig. 26(b)) with an AUC value of 1 for all classifiers and both datasets. These results demonstrate the effectiveness of the present method for correctly identifying glaucoma. Based on the AUC results, we believe that the methodology has high applicability.

We also compared our method with another classical descriptor in the literature, GLCM. This descriptor characterizes texture by jointly distributing neighboring pairs of pixels within an image, considering a particular spatial relationship [61]. This distribution forms an array containing different combinations of gray pixel values in an image. Then, we calculated the attributes of contrast, dissimilarity, homogeneity, energy, correlation and angular second moment (ASM). The results presented in Table 4 were validated considering the same scenario for the best result of the proposed method (Fig. 26(b)), i.e., tests with all channels, CLAHE, and masks. Table 4 shows the results obtained.

4.4. Analysis with other related works

We present a brief comparison between the results of our methodology and some studies cited in Section 2, which perform the segmentation of the OD and make the classification of the regions extracted from OD into glaucomatous and normal. We know that the comparison with other works is a delicate and complex task since many factors influence a reliable comparison, such as the image dataset, validation metrics, and algorithms, among others. Therefore, Tables 5 and 6 show only the relation of the results achieved in the proposed method with the results presented in the related works.

Table 5 presents a comparison between the results obtained with our method and with other methods for the OD segmentation task. Based on the results, we can affirm that our methodology presented results as relevant as the others. It is clear that the methodology used for segmentation is effective and generic, presenting promising values for the evaluation metrics.

The results presented in Table 5 show that only the works of [15,19,20] and [21] have a Dice score higher than the proposed method. However, they use less images than our approach. Although the proposed method uses a cGAN, we have built a method with a good generalization that can run on low-cost hardware and provide good results. Importantly, after the model

training, the proposed method takes on average 0.1s to segment an image. Finally, we believe that our approach also provides a solid conceptual basis for cGAN, enabling future studies.

Table 6 shows the results for the classification stage in normal and glaucomatous images. The methodology proposed by Ferreira et al. [7] stands out for achieving precision values that are shown in our proposal. However, it is important to note that our method uses a larger number of images. In addition to presenting a more effective segmentation step, our method is similar to the best recently developed studies that seek to identify glaucoma automatically.

Finally, after analyzing the comparisons presented in Tables 5 and 6, we can consider that our method is promising and that we have reached the main goal of the study, which is the classification of glaucoma through the analysis of retinal images.

4.5. Discussions

After analyzing the results presented in the previous sections, the following points should be highlighted:

- The semantic segmentation of the OD made with a cGAN capable of delimiting the ROI precisely;
- The exploration of texture through descriptors based on taxonomic diversity and taxonomic distinction indexes was shown to be efficient;
- Texture features of ROIs using all RGB channels and gray tones provide better discrimination capacity, presenting high variability among species, allowing the classifier to successfully separate healthy and glaucomatous images;
- The application of contrast enhancements is effective because it maps the luminous intensity values to similar bands among the classes of images analyzed; and,
- The application of internal and external masks favors the discovery of diversity patterns in areas near the border as well as internally.

5. Conclusion

This work presents an automatic method that uses cGANs, taxonomic diversity (Δ) and distinction indexes (Δ^*) capable of distinguishing and classifying abnormalities in retinal images that give an early indication of the presence of glaucoma, which can be used by specialists as a second opinion.

Even though OD segmentation is a complex and high-level abstraction task, the cGAN used in our method, alongside traditional image processing practices, presents a useful OD segmentation, providing robust and generic results. This work also presents ways of overcoming the limitation of a dataset containing few images for training.

The results demonstrate the promising performance of texture extraction techniques, as the proposed indexes of taxonomic diversity proved to be efficient in the characterization of OD regions. We have achieved encouraging results, showing significant

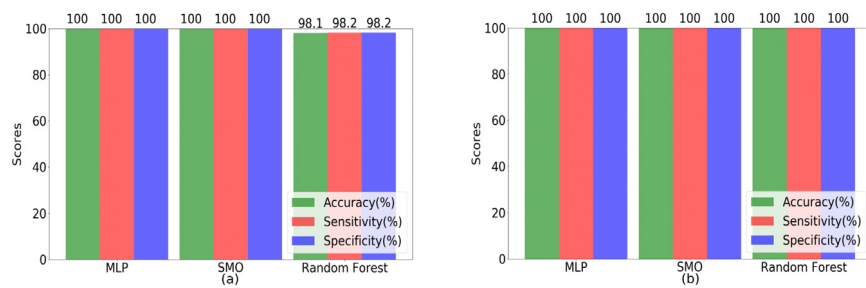


Fig. 26. (a) Tests with all channels, histograms equalization and masks, and (b) tests with all channels, CLAHE and masks.

Table 5

Comparison of the results of the segmentation stage with the related works.

Method	Image dataset	Sample	Dice score	IOU	Recall
[19]	RIM-ONE v1, DRIONS-DB, Drishti-GS	370	0.95	0.89	–
[15]	Private dataset	30	0.92	–	–
[1]	RIM-ONE and DRIONS-DB	556	0.81	–	–
[20]	RIM-ONE	159	0.98	0.96	–
[7]	RIM-ONE v1, DRIONS-DB, Drishti-GS	370	0.84	0.73	0.78
[21]	Drishti-GS	110	0.97	–	–
Proposed method	Dristh-GS and RIM-ONE	556	0.89	0.8	0.86

Table 6

Comparison of the results of the classification stage with the related works.

Method	Image dataset	Sample	Acc (%)	Spc (%)	Sen (%)	AUC
[13]	Private dataset	20	90	–	–	–
[18]	Dristh-GS, DRIVE	–	–	–	–	0.7
[14]	Private dataset and Public dataset MIAG	560	98.3	96.6	100	–
[1]	Dristh-GS and RIM-ONE	556	99.6	99.3	100	0.9
[7]	Dristh-GS, DRIONS-DB and RIM-ONE v1	370	100	100	100	1
Proposed method	Dristh-GS and RIM-ONE	556	100	100	100	1

values in all outcome metrics and a success rate of 100% using several classifiers.

Thus, we believe that the proposed method may integrate a computational tool for glaucoma detection in real scenarios. It can benefit specialists with auxiliary analysis during the glaucoma detection, which guarantees the patient better life quality and a more favorable prognosis regarding the treatment of glaucoma.

To improve the proposed method, we intend to develop the following as future work:

- Transform the proposed methodology into an application which provides a screening tool and a complementary analysis in medical scenarios;
- Augment the training and test image sets to improve the capability and generalization of the proposed method;
- Use other classifiers to evaluate the characterization capacity of the proposed indexes, for example, XGBoost [62];
- Evaluate other cGAN architectures to achieve a faster and more accurate learning phase; and,
- Extend the methodology to investigate different eye diseases.

Declaration of competing interest

No author associated with this paper has disclosed any potential or pertinent conflicts which may be perceived to have impending conflict with this work. For full disclosure statements refer to <https://doi.org/10.1016/j.asoc.2020.106165>.

CRediT authorship contribution statement

Tomaz Ribeiro Viana Bisneto: Software, Methodology, Validation, Writing - original draft. **Antonio Oseas de Carvalho Filho:**

Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing - original draft, Writing - review & editing, Supervision. **Deborah Maria Vieira Magalhães:** Validation, Formal analysis, Writing - review & editing.

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